Xinyu Weng

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Research Outline (past-future) ; *Personal-site*; *Portfolio*

EDUCATION

South China University of Technology*(U.S. News Global 162)* | M.S. in Industrial Design Engineering 2020-2023

- National Scholarship (Top 1% for academic excellence)
- Specialisms: *Affective Computing; Immersive HCI (AR); Behavior Design; Gamification; Digital Psychotherapy; Quantitative Experimental Design*

Zhengzhou University | B.S. in Packaging Engineering 2016-2020

- Overall Rank: Top 5%
- Recipient of 10+ National & Provincial Awards

RESEARCH INTERESTS

Digital Health · Human-AI Interaction · Physical AI · Affective Computing · Multimodal Perception
How multimodal perception enables agents to better understand users, and adaptively adjust their interactions via affective modeling and reasoning, delivering more personalized and healing companionship.

RESEARCH EXPERIENCE

E-PrEmo: How Visual Factors Influence on Emotion Regulation Interaction **Research Lead**
Published in Frontiers in Psychology (SSCI-Q1, IF: 3.8) Jan. 2022

- **Developed and executed *E-PrEmo method*,** contextually screening 10 emotions from a 60-emotion pool and leveraging a Rank-Based Weighting formula to map multi-layered continuous emotion values, identifying virtual scenes as the strongest driver of positive valence.
- **Proposed a strategic for emotion-regulation products**, defining the psychological roles of visual factors — scenes (relaxation & liberation), stickers (joy & surprise), data viz (satisfaction & frequent use).

SandplayAR: Evaluation of psychometric game for people with generalized anxiety disorder **Research Lead**
Published in The Arts in Psychotherapy (SSCI-Q3, IF: 1.7) Jun. 2022

- **Designed SandplayAR**, a smartphone-based AR psychometric game utilizing *2D card recognition* and *interactive 3D worlds* to facilitate accessible, self-help emotional expression for individuals with generalized anxiety disorder.
- **Formulated a mixed-methods evaluation**, integrating SIHEQ questionnaire with physiological metrics (skin conductance, blood pressure), which revealed significant improvements in *unconscious emergence* and *conscious perception activation,* and demonstrated that such immersive gamification tools can relieve tension.

HiMeet: An AI-Assisted AR Intervention System for Socially Anxious Individuals **Research Lead**
Master's Dissertation (Double A Grade, Top 5%) Jun. 2023

- **Extensive Survey and Path Analysis:** Conducted *Mediation Analysis* on questionnaire data (N=101) to examine maladaptive social behaviors in virtual and real-world contexts. Results: *Real-life Support* negatively predicted *Online Social Dependence* through the mediating role of *Social Anxiety*, especially among college students(aged 18-25), critically examined the boundaries of digital social interaction and advocate for offline engagement.
- **Individual Cognitive–Behavioral Analysis:** Conducted exposure scenarios based on a *Role-Play Assessment,* integrating subjective scales, physiological signals and behavioral observation. Results informed intervention strategies to guide product design.
- **Designed HiMeet concept demo (AI+AR):** visualized *Social Cues* (behavior/emotion recognition), delivered AI *Social Support* (feedback/tips), and designed LBS-based *Social Game* "Encounter Rose".

PUBILICATIONS AND PATENTS

- Zhiyong Xiong *(Advisor)*, **Xinyu Weng**, Yu Wei (2022) Research on the Influence of Visual Factors on Emotion Regulation Interaction. Front. Psychol. 12:772642. *Doi: 10.3389/fpsyg.2021.772642*
- Zhiyong Xiong *(Advisor)*, **Xinyu Weng**, Yu Wei, SandplayAR: Evaluation of psychometric game for people with generalized anxiety disorder,The Arts in Psychotherapy,Volume 80, 101934. *Doi: 10.1016/j.aip.2022.101934*
- Patent No. CN 215753568 U (2022), A Glue-Free Packaging Box for Rapid Prototyping (Utility Model)
- Patent No. CN 215753557 U (2022), A Recyclable Anti-Theft Express Packaging Box (Utility Model)

WORK EXPERIENCE

BYD Auto Industry Co., Ltd. Global top automaker | 20M+ users
Senior Product Manager (Recognized as High-Potential Talent) July 2023 - July 2025

SmartCharging: Conduct an Immersive Scenario to Make Wait Time More Engaging

- [CN Vision] **Revitalized a neglected product into high-marketing-value asset,** Update lifted usage(13% to 30%) and user-friendliness score (6 to 8). Debuted at the *Mega-Watt Flash Charging Event*.
- [UK Vision] **localized vision in UK** via on-site fieldwork in Manchester. Survey report laid a solid foundation for overseas strategy.

TravelAI: Multidimensional Search with an Voice-Map Agent

- Independently plan features and make practical demo. It enables destination search to go beyond keyword matching and static rules by interpreting natural language, spatial relations and urgency cues, and integrating semantic retrieval with enriched POI data.

Wangsu Science & Technology Co., Ltd. Global CDN Provider | 3 k+ Edge Nodes
Senior Product Manager July 2025 - Now

SalesAI: A Chatbot for Enterprise Efficiency (0-to-1)

- **Planned SalesAI's roadmap and designed core features**, covering before-sale opportunities acquisition, in-sale knowledge base retrieval, and after-sale information smart-management.
- **Designed a Hybrid Intent Recognition strategy**, regex-based keyword matching + LLM-driven context analysis + corpus vector retrieval + dynamic similarity thresholds, elevating the F1-score from 66.8% to 89.1%.
- **Designed UX standards**, including *Digital Avatar*, *Data Dashboard*, *Agents Store*, *ShortCuts*, etc
- **Constructed AgentWorkflows:** Designed business logic and organised data structures. Exposed 21 APIs to support NLP-triggered auto data entry and query, replacing cumbersome multi-layered UI interaction.

AWARDS AND HONORS

- National Scholarship (1%) | Ministry of Education — Jan. 2022
- Practice-Oriented Graduate Student Honor (1%) | Graduate School of SCUT — Jan.2022
- Institute 3rd Prize in BYD Innovation Proposal Contest, *Work: parking assistant product demo* — Aug. 2023
- National 3rd Prize in UXDA (International User Experience Innovation Competition) — Oct. 2021
  Work: Campus Pick-up service design
- National 1st Prize in China Packaging Creative Design Contest, *Work: Waxing and Waning* — Aug. 2019
- National 2nd & Provincial 1st Prize in China Collegiate Computing Contest (Digital Media) — Apr. 2019
  Work: Dream Shatter
- S Prize in Interdisciplinary Contest in Modeling (ICM) — Mar. 2019

SKILLS

- **Technical & Prototyping:** Figma, PS(Adobe Photoshop), Procreate; AI-assisted Coding, VS Code/Arduino IDE; Rhino, CAD; AI+MCP+Blender
- **Research Methods:** Data-Driven User Research (Psychophysiological Measurement; Behavioral Quantification)
- **Data Analysis:** Inferential Statistics (t-test, ANOVA, etc.) via Python; SPSS
- **Languages:** Chinese (Native); English (Fluent)
- **Personal Interests:** Skiing; Photography; Public Speaking & Event Hosting